

PDA Lab3 OpenROAD Tutorial

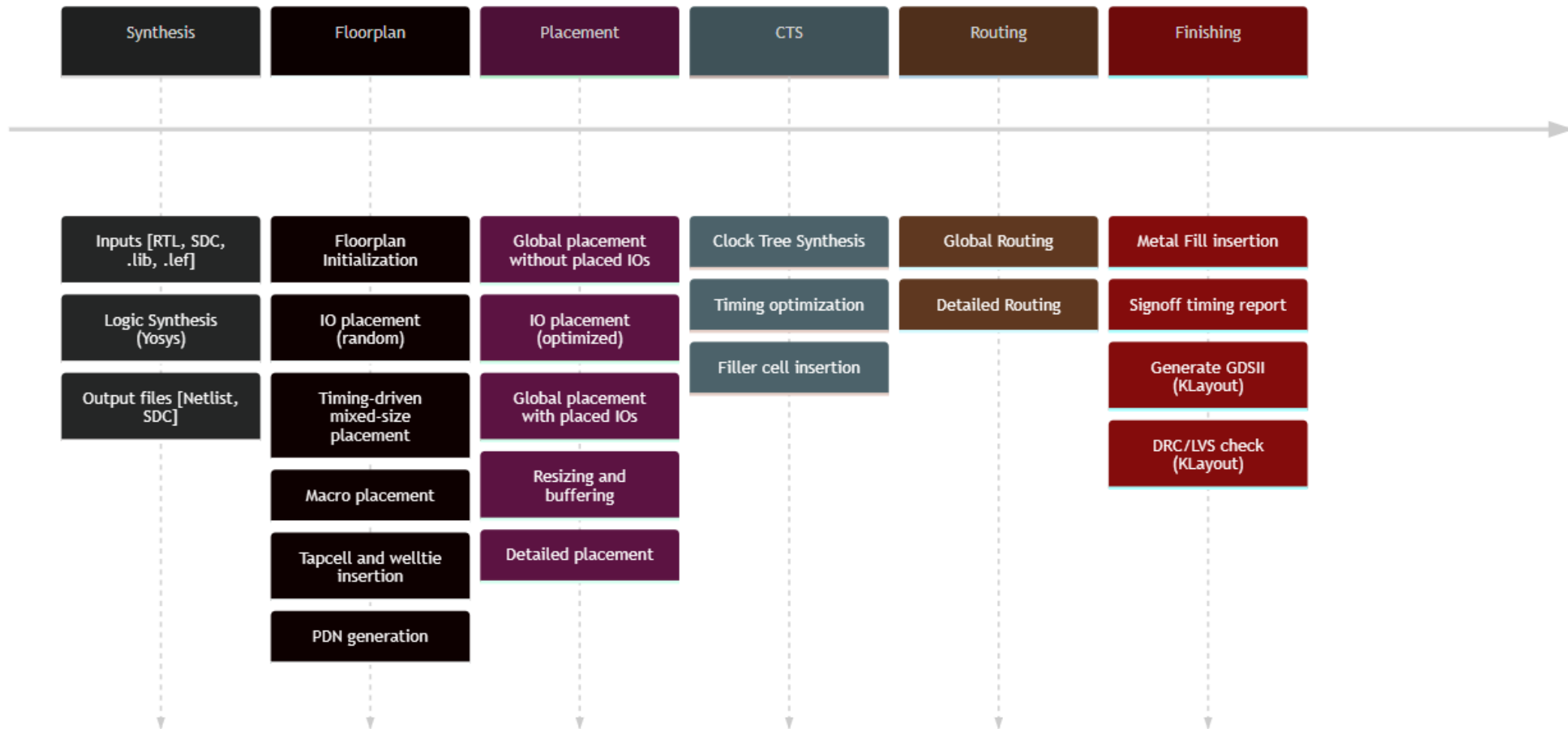
Advisor: Mark Po-Hung Lin
TA: 吳瑞宸 林酩荏

Introduction

- ❖ OpenROAD is the leading open-source, foundational application for semiconductor digital design. The OpenROAD flow delivers an Autonomous, No-Human-In-Loop (NHIL) flow, 24 hour turnaround from RTL-GDSII for rapid design exploration and physical design implementation.
- ❖ [Doc](#)
- ❖ [Github](#)

Flow

RTL-GDSII Using OpenROAD-flow-scripts



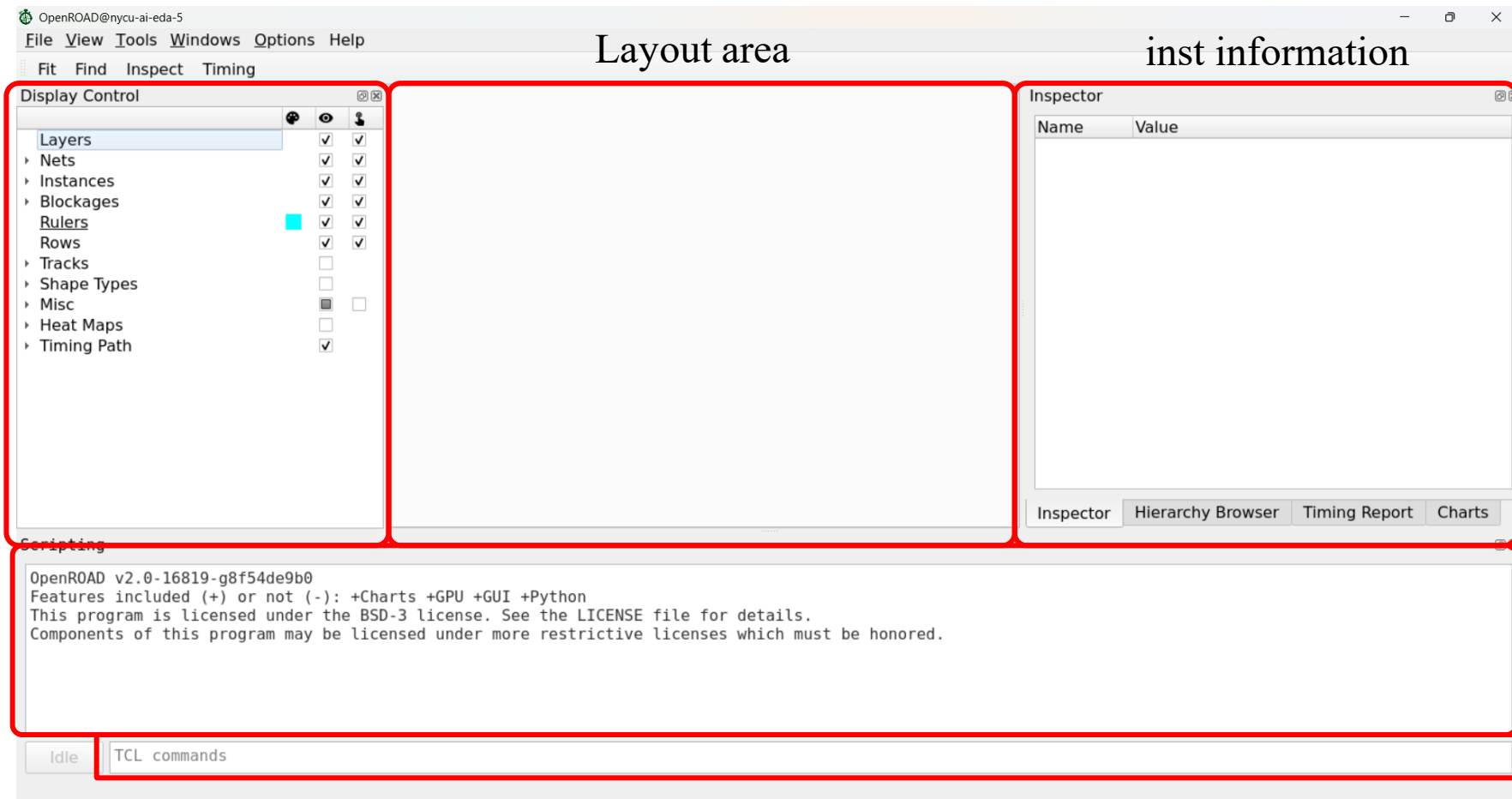
Openroad

\$ openroad -gui &

You can operate in tcl shell

Display panel

- Instances
- Rows
- Heat Maps



output log

tcl command

Load files (lef)

Scripting



```
OpenROAD v2.0-16819-g8f54de9b0
Features included (+) or not (-): +Charts +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
```

Idle

read_lef ispd19_sample.input.lef

Scripting



```
OpenROAD v2.0-16819-g8f54de9b0
Features included (+) or not (-): +Charts +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
>>> read_lef ispd19_sample.input.lef
[INFO ODB-0227] LEF file: ispd19_sample.input.lef, created 18 layers, 22 vias, 16 library cells
```

Idle

TCL commands

Load files (def)

Scripting

```
OpenROAD v2.0-16819-g8f54de9b0
Features included (+) or not (-): +Charts +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
>>> read_lef ispd19_sample.input.lef
[INFO ODB-0227] LEF file: ispd19_sample.input.lef, created 18 layers, 22 vias, 16 library cells
```

Idle

read_def ispd19_sample.input.def

The screenshot shows the OpenROAD GUI with the following components:

- Menu Bar:** File, View, Tools, Windows, Options, Help
- Display Control:** A panel on the left showing a list of layers and vias with checkboxes and color indicators.

Layer/Via	Color	Check 1	Check 2
Other		☐	☐
Metal1	Blue	☒	☒
Via1	Blue	☒	☒
Metal2	Red	☒	☒
Via2	Red	☒	☒
Metal3	Green	☒	☒
Via3	Green	☒	☒
Metal4	Yellow	☒	☒
Via4	Yellow	☒	☒
Metal5	Pink	☒	☒
Via5	Pink	☒	☒
Metal6	Cyan	☒	☒
Via6	Cyan	☒	☒
Metal7	Purple	☒	☒
Via7	Purple	☒	☒
Metal8	Purple	☒	☒
Via8	Purple	☒	☒
Metal9	Olive	☒	☒
- Design View:** A central window showing a top-down view of the design, which appears as a grid of red and green rectangles on a black background.
- Inspector:** A panel on the right with a table for Name and Value, and tabs for Inspector, Hierarchy Browser, Timing Report, and Charts.
- Scripting Console:** A panel at the bottom showing the execution of commands and their output.

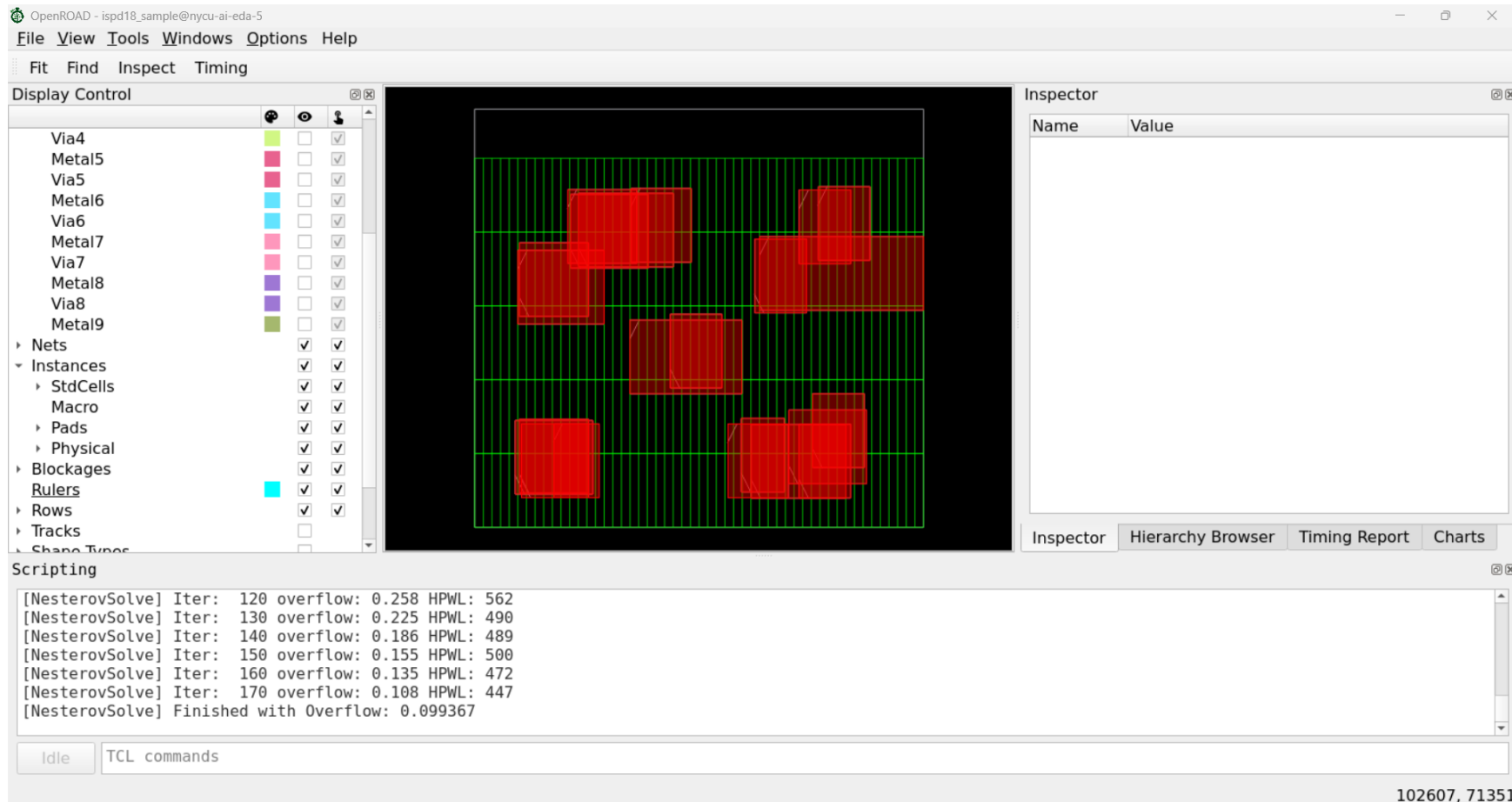
```
>>> read_lef ispd19_sample.input.lef
[INFO ODB-0227] LEF file: ispd19_sample.input.lef, created 18 layers, 22 vias, 16 library cells
>>> read_def ispd19_sample.input.def
[INFO ODB-0127] Reading DEF file: ispd19_sample.input.def
[INFO ODB-0128] Design: ispd18_sample
[INFO ODB-0131] Created 22 components and 146 component-terminals.
[INFO ODB-0133] Created 11 nets and 22 connections.
[INFO ODB-0134] Finished DEF file: ispd19_sample.input.def
```

The status bar at the bottom right shows the coordinates 86369, 82279.

global_placement

\$ global_placement -density <value>

Value recommend from 0.8 to 0.95 depends on case



extract

\$ source extract.tcl

Filename: designName_insts.gp

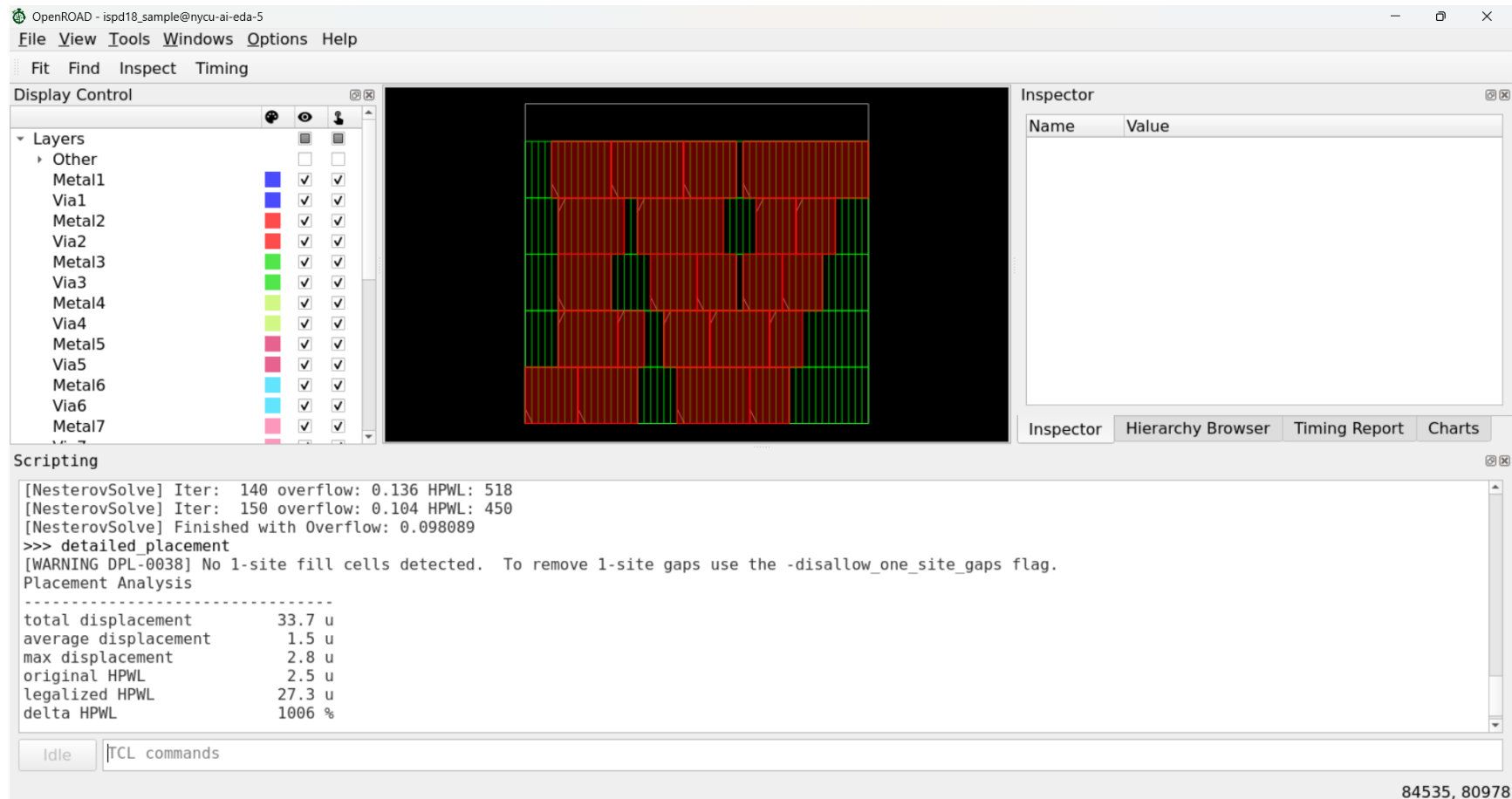
```
DBU_Per_Micron 2000
DieArea_LL 83600 71820
DieArea_UR 104400 91200
Site_Width 400
Site_Height 3420

Name LLX LLY Width Height Type
inst2015 85600 78660 3200 3420 CELL
inst2591 88800 85500 4400 3420 CELL
inst2908 94800 82080 2800 3420 CELL
...
```


detail_placement & legalization

\$ detail_placement

OpenROAD would have some statistics.



The screenshot shows the OpenROAD GUI with the following components:

- Display Control:** A list of layers on the left, including Metal1 through Metal7 and Via1 through Via7, with checkboxes for visibility and color selection.
- Inspector:** A panel on the right with a table for Name and Value, and tabs for Inspector, Hierarchy Browser, Timing Report, and Charts.
- Scripting:** A window at the bottom showing the output of the `detail_placement` command.

The scripting window output is as follows:

```
[NesterovSolve] Iter: 140 overflow: 0.136 HPWL: 518
[NesterovSolve] Iter: 150 overflow: 0.104 HPWL: 450
[NesterovSolve] Finished with Overflow: 0.098089
>>> detail_placement
[WARNING DPL-0038] No 1-site fill cells detected. To remove 1-site gaps use the -disallow_one_site_gaps flag.
Placement Analysis
-----
total displacement      33.7 u
average displacement    1.5 u
max displacement        2.8 u
original HPWL           2.5 u
legalized HPWL          27.3 u
delta HPWL              1006 %
```

The status bar at the bottom shows "Idle" and "TCL commands". The bottom right corner of the window displays the coordinates "84535, 80978".

Cell information

- ❖ The inspector shows the information such as inst_name, status, orientation, bounding box, width, height.



The image shows a circuit design tool interface. On the left, a grid of cells is displayed, with a yellow box highlighting a specific cell. On the right, an 'Inspector' window displays the properties of the selected cell.

Name	Value
Type	Inst
Name	inst7234
Block	ispd18_sample
Module	<top>
Master	AO22XL
Description	Standard cell
Placement status	PLACED
Source type	NONE
Dont Touch	False
Orientation	R0
X	92800
Y	85500
Items	7 items
Timing/Power	inst7234
BBox	(92800, 85500), (96000, 88920)
BBox Width, Height	(3200, 3420)

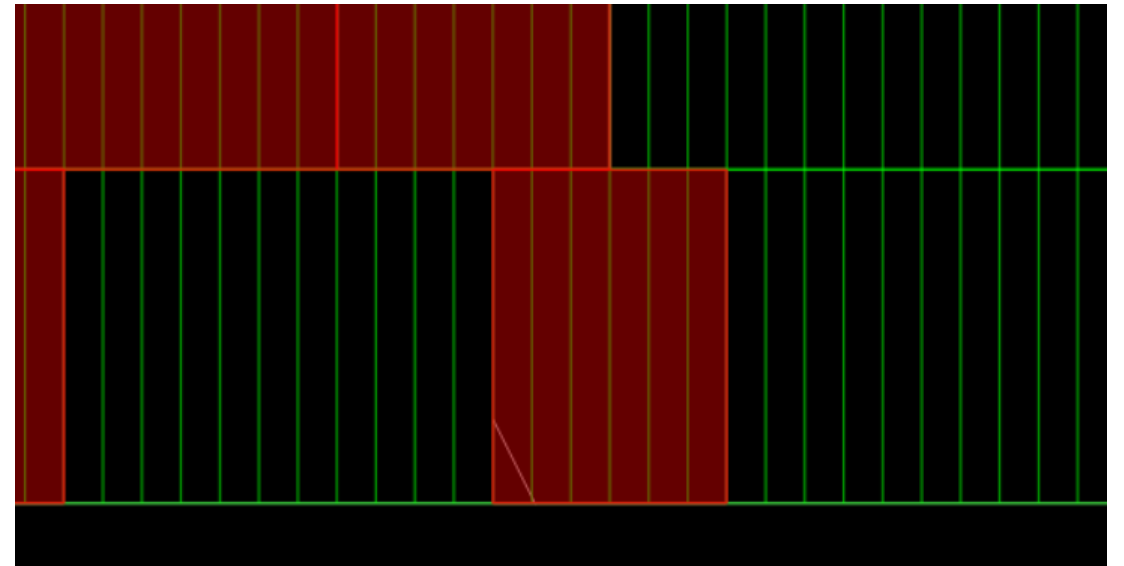
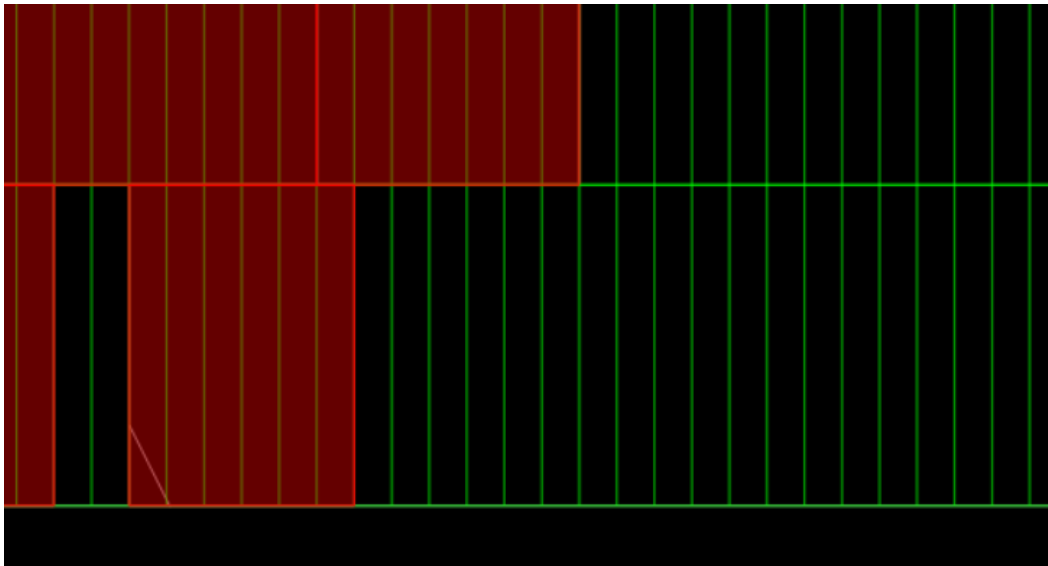
Inspector Hierarchy Browser Timing Report Charts

place_cell

```
>>> place_cell -inst_name inst6050 -orient R0 -origin {48.6 35.91}
```

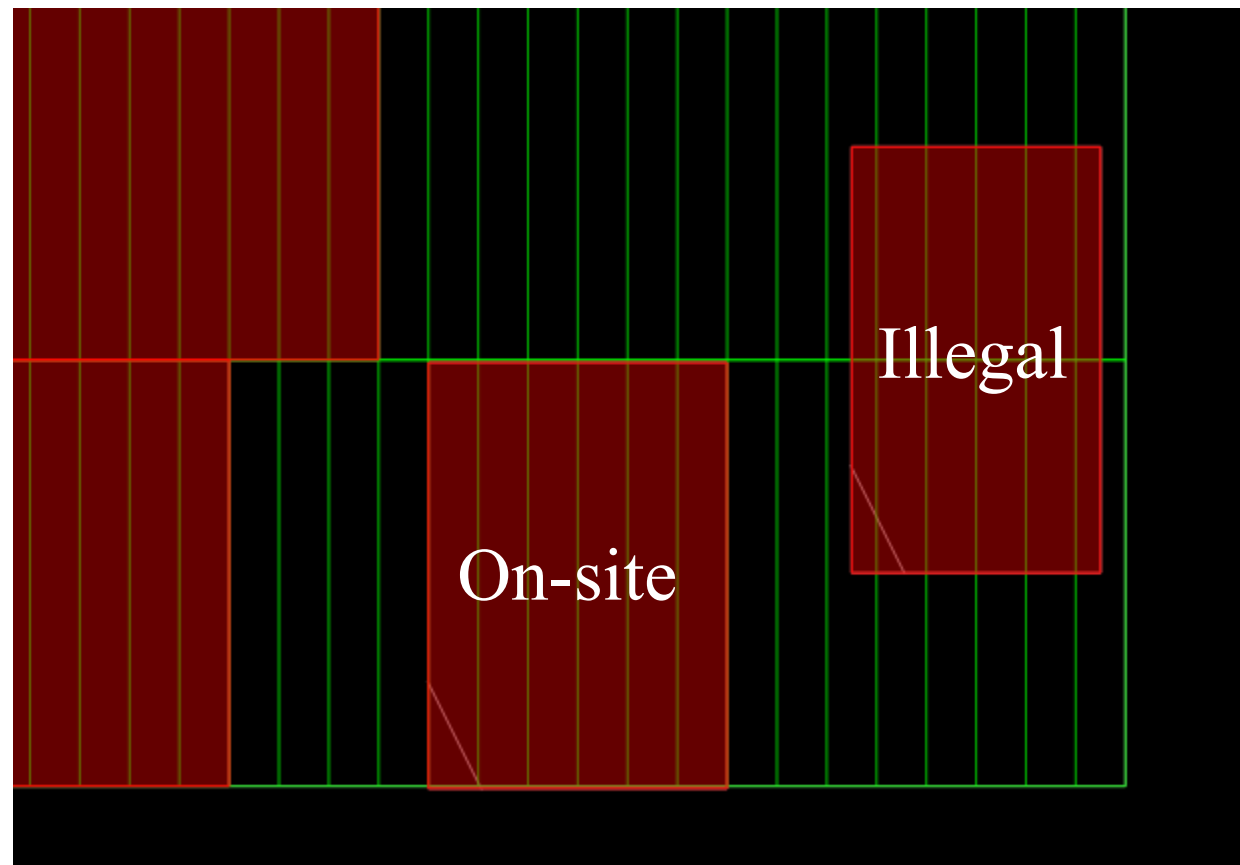
Idle

TCL commands



Legality

- ❖ On-site: Cells must be placed with their bottom-left corner aligned to the site grid defined by the input design data.



Legality check

\$ `check_placement -verbose` # return 0 if pass

The screenshot displays the OpenROAD software interface for a project named 'ispd18_sample@nycu-ai-eda-5'. The main window shows a top-level view of a circuit layout with various colored regions representing different layers or components. A central yellow/orange rectangular area is highlighted, indicating the location of the failed placement check.

The **Display Control** panel on the left lists various layers and nets, each with a color-coded square and checkboxes for visibility. The **Inspector** panel on the right shows the details of the selected instance, `inst4189`, which is a standard cell of type `AOI22X2`. The placement status is `PLACED`, and the bounding box is `(93000, 71820), (97400, 75240)`.

The **Scripting** panel at the bottom shows the command `check_placement -verbose` being executed. The output indicates that the detailed placement checks failed for instance `inst4189`, with error messages `[ERROR DPL-0033] detailed placement checks failed.` and `[ERROR GUI-0070] DPL-0033`. A warning message `[WARNING DPL-0006] Site aligned check failed (1).` is also present.

The status bar at the bottom shows the current instance `inst4189` and the coordinates `100970, 74032`.

Heatmap

OpenROAD - ispd18_sample@nycu-ai-eda-5

File View Tools Windows Options Help

Fit Find Ruler K

Display Con Heat maps

Global connect Ctrl+G

Metal9

Nets

Instances

Blockages

Rulers

Rows

Tracks

Shape Types

Misc

Heat Maps

Power Density

Estimated Congestion (RUDY)

Routing Congestion

IR Drop

Placement Density

Power Density

Routing Congestion

Estimated Congestion (RUDY)

IR Drop

Timing Path

Scripting

Idle TCL commands

Power Density

Estimated Congestion (RUDY)

Routing Congestion

IR Drop

Placement Density

Power Density

Routing Congestion

Estimated Congestion (RUDY)

IR Drop

Timing Path

Scripting

total displacement 0.1 u

average displacement 0.0 u

max displacement 0.1 u

original HPWL 33.1 u

legalized HPWL 33.0 u

delta HPWL 0 %

Scripting

total displacement 0.1 u

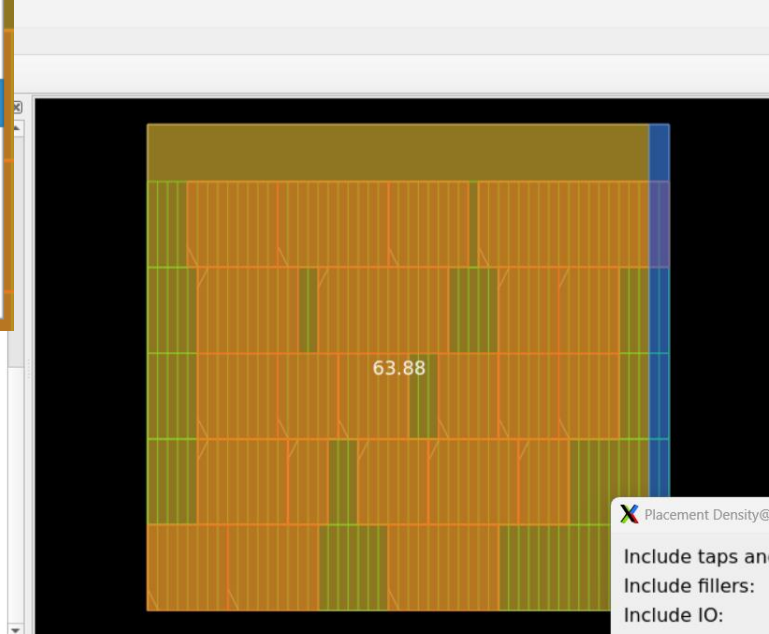
average displacement 0.0 u

max displacement 0.1 u

original HPWL 33.1 u

legalized HPWL 33.0 u

delta HPWL 0 %



Inspector

Name Value

Type Inst

Name inst4189

Block ispd18_sample

Module <top>

Master AOI22X2

Description Standard cell

Placement status PLACED

Source type NONE

Dont Touch False

Orientation R0

X 93000

Y 71820

ITerms 7 items

Timing/Power inst4189

BBox (93000, 71820), (97400, 75240)

Inspector

Name Value

Type Inst

Name inst4189

Block ispd18_sample

Placement Density@nycu-ai-eda-5

Include taps and endcaps: ☒

Include fillers: ☐

Include IO: ☐

Log scale ☐

Reverse Log scale ☐

Show numbers ☒

Show legend ☐

Grid X 10.00 μm Y 10.00 μm

Minimum 0.000 % Show values below ☐

Maximum 100.000 % Show values above ☒

Color alpha 150

Rebuild data Close

Heatmap Density

\$ gui::dump_heatmap "Placement" "<path to csv>"

```
Scripting
average displacement      0.0 u
max displacement         0.1 u
original HPWL            33.1 u
legalized HPWL           33.0 u
delta HPWL               0 %

>>> gui::dump_heatmap "Placement" "stat.csv"
```

Idle TCL commands

x0	y0	x1	y1	value
41.8	35.91	51.8	45.6	63.88235
51.8	35.91	52.2	45.6	17.64706

Note

- ❖ flow.tcl # Overall flow wrapper
- ❖ extract.tcl # Extract input.gp
- ❖ Please use mobaxterm to connect to the server, **DO NOT** use vscode ssh
 - Server ip:140.113.30.188:7405
 - Account: same
 - Password: nycu2026